Lecture 20

Real Forms and Noncompact Structure

Objectives. *A connected real semisimple group with finite centre has a product decomposition into a compact subgroup, an abelian subgroup and a nilpotent subgroup. This lecture compares compact and split real forms and states the Cartan and Iwasawa decompositions. It proves both decompositions for* $\mathrm{SL}(2,\mathbb{R})$*, where they give coordinates on the upper half-plane.*

By the end of this lecture you will be able to:

- Distinguish the compact and split examples of real forms using their Killing forms and Cartan involutions.
- State the Cartan and Iwasawa decompositions with all their hypotheses.
- Compute $g = kan$ and $g = k \exp X$ for an explicit matrix in $\mathrm{SL}(2,\mathbb{R})$.
- Read both decompositions as coordinates on the upper half-plane.

20.1 Real forms and Cartan involutions

Throughout this section $\mathfrak{g}_{\mathbb{C}}$ is a complex semisimple Lie algebra with Killing form

$$\kappa(X,Y) = \operatorname{tr}(\operatorname{ad} X \operatorname{ad} Y).$$

We cite Knapp, *Lie Groups Beyond an Introduction*, digital second edition 2023, as Knapp Beyond.

Definition 20.1. A *real form* of $\mathfrak{g}_{\mathbb{C}}$ is a real Lie subalgebra $\mathfrak{g} \subset \mathfrak{g}_{\mathbb{C}}$ with $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g} \oplus i\mathfrak{g}$ as real vector spaces.

A real basis of $\mathfrak{g}$ is then a complex basis of $\mathfrak{g}_{\mathbb{C}}$. For $X \in \mathfrak{g}$, the matrix of $\operatorname{ad} X$ in this basis is real. It is the same over $\mathbb{R}$ and over $\mathbb{C}$. Hence the Killing form of $\mathfrak{g}$ is the restriction $\kappa|_{\mathfrak{g}}$. It is real and nondegenerate. Its signature is unchanged by real isomorphisms.

Definition 20.2. A real form $\mathfrak{g}$ is *compact* if $\kappa|_{\mathfrak{g}}$ is negative definite. It is *split* if it contains an abelian subalgebra $\mathfrak{h}_0$ with the following two properties. Every $\operatorname{ad} H$ with $H \in \mathfrak{h}_0$ is diagonalisable over $\mathbb{R}$. The sum $\mathfrak{h}_0 \oplus i\mathfrak{h}_0$ is a Cartan subalgebra of $\mathfrak{g}_{\mathbb{C}}$.

The Lie Algebras course supplies both types [I]. The real span of a Chevalley basis is a split form. Weyl's unitary trick turns the same basis into a compact form; see Knapp Beyond, Section VI.1.

The name “compact” refers to compact groups. By Lecture 6, the Lie algebra of a compact group has an Ad-invariant inner product, so each $\operatorname{ad} X$ is skew and

$$\kappa(X,X) = \operatorname{tr}\left((\operatorname{ad} X)^2\right) \le 0.$$

So a real form whose Killing form takes a positive value is not the Lie algebra of a compact group.

Example 20.3 (The two forms of $\mathfrak{sl}(2,\mathbb{C})$)**.** On $\mathfrak{sl}(2,\mathbb{C})$ one has $\kappa(X,Y) = 4\operatorname{tr}(XY)$. Take

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Then $H^2 = I$, $(E+F)^2 = I$ and $(E-F)^2 = -I$. The products

$$H(E+F) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad H(E-F) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad (E-F)(E+F) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

are traceless. In the basis $(H, E+F, E-F)$ of $\mathfrak{sl}(2,\mathbb{R})$ the Gram matrix of κ is therefore $\operatorname{diag}(8,8,-8)$. The signature is $(2,1)$. The real form $\mathfrak{sl}(2,\mathbb{R})$ is split with $\mathfrak{h}_0 = \mathbb{R}H$, because $\operatorname{ad} H$ has eigenvalues $2, 0, -2$ on E, H, F.

The course basis of $\mathfrak{su}(2)$ is

$$A = iH, \qquad B = E - F, \qquad C = i(E+F).$$

Thus $\kappa(A,A) = i^2 \cdot 8 = -8$, $\kappa(C,C) = -8$ and $\kappa(B,B) = -8$. The cross terms are multiples of the vanishing ones above. The Gram matrix is $-8I$, so $\mathfrak{su}(2)$ is compact. The signatures $(2,1)$ and $(0,3)$ differ, so the two real forms are not isomorphic.

The two real forms in the example share the line $\mathbb{R}B$. The split form adds the plane $\operatorname{span}(H, E+F)$. The compact form adds the plane $i\operatorname{span}(H, E+F)$, spanned by A and C. Thus

$$\mathfrak{sl}(2,\mathbb{R}) = \mathbb{R}B \oplus \operatorname{span}(H, E+F), \qquad \mathfrak{su}(2) = \mathbb{R}B \oplus i\operatorname{span}(H, E+F).$$

A Cartan involution makes this relation precise. Let $\mathfrak{g}$ now be any real semisimple Lie algebra. Write $\kappa(X,Y) = \operatorname{tr}(\operatorname{ad} X \operatorname{ad} Y)$ for the Killing form of $\mathfrak{g}$ itself. For a real form of $\mathfrak{g}_{\mathbb{C}}$, this is the restriction $\kappa|_{\mathfrak{g}}$ described above.

Definition 20.4. A *Cartan involution* of $\mathfrak{g}$ is a Lie algebra automorphism θ with $\theta^2 = \mathrm{id}$ such that

$$\kappa_\theta(X,Y) = -\kappa(X,\theta Y)$$

is positive definite on $\mathfrak{g}$.

The form κ_θ is symmetric. Since θ is an automorphism, $[\theta X, Z] = \theta[X, \theta^{-1}Z]$ for all $Z \in \mathfrak{g}$. So $\operatorname{ad}(\theta X) = \theta \operatorname{ad} X\, \theta^{-1}$. Hence $\kappa(\theta X, \theta Y) = \operatorname{tr}(\theta \operatorname{ad} X \operatorname{ad} Y\, \theta^{-1}) = \kappa(X,Y)$. Using $\theta^2 = \mathrm{id}$ and the symmetry of κ, we get

$$\kappa(X,\theta Y) = \kappa(\theta X, \theta^2 Y) = \kappa(\theta X, Y) = \kappa(Y, \theta X).$$

Thus $\kappa_\theta(X,Y) = \kappa_\theta(Y,X)$. So κ_θ is an inner product on $\mathfrak{g}$, as the spectral theorem in § 20.2 requires.

Write $\mathfrak{k}$ and $\mathfrak{p}$ for the $(+1)$- and (-1)-eigenspaces of θ. Since $\theta^2 = \mathrm{id}$, we have $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$. Because θ is an automorphism,

$$[\mathfrak{k},\mathfrak{k}] \subset \mathfrak{k}, \qquad [\mathfrak{k},\mathfrak{p}] \subset \mathfrak{p}, \qquad [\mathfrak{p},\mathfrak{p}] \subset \mathfrak{k}.$$

For instance, $X, Y \in \mathfrak{p}$ give $\theta[X,Y] = [-X,-Y] = [X,Y]$. Every real semisimple Lie algebra has a Cartan involution, unique up to inner automorphism [C: Knapp Beyond, Section VI.2].

Proposition 20.5. *Let $\mathfrak{g} = \mathfrak{sl}(n,\mathbb{R})$ and $\theta(X) = -X^T$.*

(a) θ is a Cartan involution. Here $\mathfrak{k} = \mathfrak{so}(n)$, and $\mathfrak{p}$ consists of the real symmetric traceless matrices.

(b) $\mathfrak{k} \oplus i\mathfrak{p} = \mathfrak{su}(n)$, and $\mathfrak{su}(n)$ is a compact real form of $\mathfrak{sl}(n,\mathbb{C})$.

(c) The traceless diagonal matrices form a maximal abelian subspace $\mathfrak{a} \subset \mathfrak{p}$. Each $\operatorname{ad} H$, $H \in \mathfrak{a}$, is diagonalisable over $\mathbb{R}$. Hence $\mathfrak{sl}(n,\mathbb{R})$ is split.

Proof. (a) Clearly $\theta^2 = \mathrm{id}$. It is an automorphism because

$$[\theta X, \theta Y] = X^T Y^T - Y^T X^T = (YX - XY)^T = \theta[X,Y].$$

On $\mathfrak{sl}(n,\mathbb{C})$ we import $\kappa(X,Y) = 2n \operatorname{tr}(XY)$. Then

$$\kappa_\theta(X,X) = 2n \operatorname{tr}(XX^T) = 2n \sum_{i,j} X_{ij}^2 > 0 \qquad (X \neq 0).$$

The eigenspace descriptions are immediate.

(b) Write $Z \in \mathfrak{gl}(n,\mathbb{C})$ as $Z = X + iY$ with X, Y real. Then $Z^* = X^T - iY^T$. So $Z^* = -Z$ exactly when $X^T = -X$ and $Y^T = Y$. Also $\operatorname{tr} Z = 0$ exactly when $\operatorname{tr} Y = 0$, since $\operatorname{tr} X = 0$ already. Hence $\mathfrak{su}(n) = \mathfrak{k} \oplus i\mathfrak{p}$. It is a real form because

$$\mathfrak{sl}(n,\mathbb{C}) = (\mathfrak{k} \oplus \mathfrak{p}) \oplus i(\mathfrak{k} \oplus \mathfrak{p}) = (\mathfrak{k} \oplus i\mathfrak{p}) \oplus i(\mathfrak{k} \oplus i\mathfrak{p}).$$

For nonzero $Z \in \mathfrak{su}(n)$, we get $\kappa(Z,Z) = 2n \operatorname{tr}(Z^2) = -2n \operatorname{tr}(Z^*Z) < 0$.

(c) Let $H = \operatorname{diag}(h_1, \dots, h_n)$ and let E_{ij} be the matrix units. Then

$$[H, E_{ij}] = (h_i - h_j) E_{ij}.$$

So $\operatorname{ad} H$ is diagonal, with real eigenvalues, in the basis of all E_{ij} with $i \neq j$ together with a basis of $\mathfrak{a}$. For maximality, choose $H \in \mathfrak{a}$ with distinct entries. If $Y \in \mathfrak{p}$ commutes with H, then $(h_i - h_j) Y_{ij} = 0$, so Y is diagonal. Finally, $\mathfrak{a} \oplus i\mathfrak{a}$ is the diagonal Cartan subalgebra of $\mathfrak{sl}(n,\mathbb{C})$. ∎

The recipe $\mathfrak{k} \oplus i\mathfrak{p}$ from the compact-form statement in Proposition 20.5 works in general. If θ is a Cartan involution of $\mathfrak{g}$, then $\mathfrak{k} \oplus i\mathfrak{p}$ is a compact real form of the complexification $\mathfrak{g}_{\mathbb{C}}$ of $\mathfrak{g}$, the complex Lie algebra in which $\mathfrak{g}$ is a real form [C: Knapp Beyond, Sections VI.1–VI.2]. For $\mathfrak{g} = \mathfrak{sl}(2,\mathbb{R})$ this recipe returns the decomposition of $\mathfrak{su}(2)$ displayed above.

20.2 Restricted roots and the global decompositions

Fix a real semisimple Lie algebra $\mathfrak{g}$ with Cartan involution θ, and write $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$.

Lemma 20.6. *For $X \in \mathfrak{p}$, the operator $\operatorname{ad} X$ is symmetric for κ_θ.*

Proof. Since θ is an automorphism and $\theta X = -X$, we have $[X, \theta Z] = -\theta[X, Z]$. Invariance of κ gives $\kappa([X,Y],W) = -\kappa(Y,[X,W])$. Hence

$$\kappa_\theta([X,Y],Z) = \kappa(Y,[X,\theta Z]) = -\kappa(Y,\theta[X,Z]) = \kappa_\theta(Y,[X,Z]).$$

∎

Choose a maximal abelian subspace $\mathfrak{a} \subset \mathfrak{p}$. The operators $\operatorname{ad} H$, $H \in \mathfrak{a}$, commute and are κ_θ-symmetric. By the spectral theorem they are simultaneously diagonalisable over $\mathbb{R}$.

Definition 20.7. For $\lambda \in \mathfrak{a}^*$ put

$$\mathfrak{g}_\lambda = \{X \in \mathfrak{g} : [H,X] = \lambda(H)X \text{ for all } H \in \mathfrak{a}\}.$$

A *restricted root* is a nonzero λ with $\mathfrak{g}_\lambda \neq 0$. The *real rank* of $\mathfrak{g}$ is $\dim \mathfrak{a}$.

The real rank does not depend on the choices of θ and $\mathfrak{a}$ [C: Knapp Beyond, Chapter VI]. First fix θ. Any two maximal abelian subspaces of $\mathfrak{p}$ are then conjugate by a finite product of automorphisms $e^{\operatorname{ad} Z}$ with $Z \in \mathfrak{k}$. This conjugacy is a statement about the Lie algebra $\mathfrak{g}$ alone.

Next, let θ' be a second Cartan involution of $\mathfrak{g}$, with (-1)-eigenspace $\mathfrak{p}'$. By the uniqueness statement in § 20.1, $\theta' = \varphi\theta\varphi^{-1}$ for an inner automorphism φ. If $\theta X = -X$, then $\theta'(\varphi X) = \varphi\theta X = -\varphi X$. Symmetrically, $\theta = \varphi^{-1}\theta'\varphi$, so $\varphi^{-1}(\mathfrak{p}') \subseteq \mathfrak{p}$. Thus φ maps $\mathfrak{p}$ onto $\mathfrak{p}'$. Since φ is a Lie algebra automorphism, it maps maximal abelian subspaces of $\mathfrak{p}$ to maximal abelian subspaces of $\mathfrak{p}'$. Hence $\dim \mathfrak{a}$ is unchanged.

Next fix $H_0 \in \mathfrak{a}$ with $\lambda(H_0) \neq 0$ for the finitely many restricted roots. Such an H_0 exists, because $\mathfrak{a}$ is not a union of finitely many hyperplanes $\ker \lambda$. Call λ positive if $\lambda(H_0) > 0$, and set

$$\mathfrak{n} = \bigoplus_{\lambda(H_0)>0} \mathfrak{g}_\lambda.$$

The Jacobi identity gives $[\mathfrak{g}_\lambda, \mathfrak{g}_\mu] \subset \mathfrak{g}_{\lambda+\mu}$. Hence $\mathfrak{n}$ is a subalgebra. Each $\operatorname{ad} X$ with $X \in \mathfrak{n}$ is nilpotent, because it strictly raises the value at H_0.

For $\mathfrak{su}(n)$ the identity is a Cartan involution, since $\kappa_{\mathrm{id}} = -\kappa$ is positive definite. The two forms of $\mathfrak{sl}(n,\mathbb{C})$ compare as follows.

Real form	θ	$\mathfrak{k}$	$\mathfrak{p}$	real rank
$\mathfrak{su}(n)$, compact	id	$\mathfrak{su}(n)$	0	0
$\mathfrak{sl}(n,\mathbb{R})$, split	$X \mapsto -X^T$	$\mathfrak{so}(n)$	symmetric traceless	$n-1$

For $\mathfrak{sl}(n,\mathbb{R})$, with $H_0 = \operatorname{diag}(n-1, n-3, \dots, 1-n)$, the algebra $\mathfrak{n}$ is the strictly upper triangular matrices (Exercise 20.2).

Theorem 20.8 (Cartan and Iwasawa decompositions; cited)**.** *Let G be a connected semisimple Lie group with finite centre. Let θ be a Cartan involution of $\mathfrak{g}$, with $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$, and let K be the connected subgroup with Lie algebra $\mathfrak{k}$. Choose $\mathfrak{a}$ and $\mathfrak{n}$ as above, and put $A = \exp \mathfrak{a}$ and $N = \exp \mathfrak{n}$. Then:*

(a) K is compact, and $K \times \mathfrak{p} \to G$, $(k, X) \mapsto k \exp X$, is a diffeomorphism;

(b) A and N are closed subgroups, and $K \times A \times N \to G$, $(k,a,n) \mapsto kan$, is a diffeomorphism.

For proofs see Knapp Beyond, Sections VI.3 and VI.4, or P. Etingof, *Lie Groups and Lie Algebras*, MIT 18.755 lecture notes, Lectures 15 and 18. The finite-centre hypothesis ensures that K is compact.

20.3 The rank-one model $\mathrm{SL}(2,\mathbb{R})$

Let $G = \mathrm{SL}(2,\mathbb{R})$ and $\theta(X) = -X^{T}$. By the description of $\mathfrak{sl}(n,\mathbb{R})$ in Proposition 20.5, $\mathfrak{k} = \mathbb{R}(E-F)$ and $\mathfrak{p} = \operatorname{span}(H, E+F)$. Take $\mathfrak{a} = \mathbb{R}H$. The restricted roots are $\pm\lambda_1$ with $\lambda_1(H) = 2$, and $\mathfrak{g}_{\lambda_1} = \mathbb{R}E$ and $\mathfrak{g}_{-\lambda_1} = \mathbb{R}F$. Hence the real rank is one. With $H_0 = H$ we get $\mathfrak{n} = \mathbb{R}E$. The subgroups K, A and N consist of the matrices

$$k_\varphi = \begin{pmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix}, \qquad a_t = \begin{pmatrix} e^{t} & 0 \\ 0 & e^{-t} \end{pmatrix}, \qquad n_x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix},$$

with $K = \mathrm{SO}(2) = \{k_\varphi\}$, $A = \{a_t\}$ and $N = \{n_x\}$. We now prove the conclusions of the Cartan and Iwasawa theorem, Theorem 20.8, directly for G, without using that theorem.

Proposition 20.9 (Iwasawa decomposition of $\mathrm{SL}(2,\mathbb{R})$). *The map $K\times A\times N \to G$, $(k,a,n)\mapsto kan$, is a diffeomorphism. Explicitly, for*

$$g = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in G, \qquad r = \sqrt{\alpha^2+\gamma^2},$$

one has $g = k_\varphi a_t n_x$ with

$$\cos\varphi = \frac{\alpha}{r}, \qquad \sin\varphi = \frac{\gamma}{r}, \qquad e^{t} = r, \qquad x = \frac{\alpha\beta+\gamma\delta}{r^{2}}.$$

Proof. The first column of g is nonzero, so $r > 0$. Let $k = k_\varphi$ with φ as stated. This is Gram–Schmidt applied to the columns of g. Using $\alpha\delta - \beta\gamma = 1$,

$$k^{-1}g = \frac{1}{r}\begin{pmatrix} \alpha & \gamma \\ -\gamma & \alpha \end{pmatrix}\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} r & \dfrac{\alpha\beta+\gamma\delta}{r} \\ 0 & \dfrac{1}{r} \end{pmatrix}.$$

On the other hand,

$$a_t n_x = \begin{pmatrix} e^{t} & e^{t}x \\ 0 & e^{-t} \end{pmatrix}.$$

These agree for $e^{t} = r$ and $x = (\alpha\beta+\gamma\delta)/r^{2}$. This proves existence.

For uniqueness, suppose $kan = k'a'n'$. Then $k'^{-1}k = a'n'(an)^{-1}$ is orthogonal and upper triangular with positive diagonal. Its first column is $(p,0)^{T}$ with $p>0$ and unit length, so $p = 1$. The second column is a unit vector orthogonal to $(1,0)^{T}$ with positive second entry, so it is $(0,1)^{T}$. Hence $k = k'$ and $an = a'n'$. Comparing diagonals gives $a = a'$, and then $n = n'$.

The multiplication map is smooth. Its inverse is given by the formulas above, which are smooth in the entries of g because $r > 0$. $\blacksquare$

In particular G is connected and diffeomorphic to a circle times $\mathbb{R}^2$. So $\mathrm{SL}(2,\mathbb{R})$ satisfies the hypotheses of the Cartan and Iwasawa theorem, Theorem 20.8, with centre $\{\pm I\}$.

For the Cartan decomposition we need the exponential on $\mathfrak{p}$. Define the entire functions

$$C(u)=\sum_{k\ge 0}\frac{u^k}{(2k)!},\qquad S(u)=\sum_{k\ge 0}\frac{u^k}{(2k+1)!}.$$

For $u=s^2\ge 0$ they are $\cosh s$ and $\sinh s/s$, with the latter interpreted as 1 at $s=0$. Let $\mathcal{P}$ be the set of symmetric positive definite 2×2 matrices of determinant 1.

Lemma 20.10. *For $Y\in\mathfrak{p}$, set $u=-\det Y\ge 0$. Then $Y^2=uI$ and*

$$\exp Y=C(u)I+S(u)Y.$$

The map $\exp\colon \mathfrak{p}\to\mathcal{P}$ is a bijection. Its inverse is

$$L(Q)=\frac{Q-cI}{S(C^{-1}(c))},\qquad c=\tfrac12\operatorname{tr}Q,$$

which is smooth on a neighbourhood of $\mathcal{P}$ among symmetric matrices.

Proof. Write $Y=\begin{pmatrix}p & q\\ q & -p\end{pmatrix}$. Then $Y^2=(p^2+q^2)I=uI$. Splitting the exponential series into even and odd powers gives the formula. The matrix $\exp Y=(\exp\frac{Y}{2})^2$ is the square of an invertible symmetric matrix. Hence $\exp Y$ is positive definite. Moreover, $\det\exp Y=e^{\operatorname{tr}Y}=1$.

Every coefficient of the series for C' is positive, so $C'>0$ on $(0,\infty)$. Also, $C'(0)=\frac12$. So $C'>0$ on some interval $(-\varepsilon,\infty)$. Thus C is a strictly increasing bijection $[0,\infty)\to[1,\infty)$, with smooth inverse near $[1,\infty)$. Also $S>0$ near $[0,\infty)$.

For injectivity, let $Q=\exp Y$. Then $\operatorname{tr}Q=2C(u)$, which determines u because C is strictly increasing on $[0,\infty)$. The formula for $\exp Y$ then gives $Y=(Q-C(u)I)/S(u)=L(Q)$.

For surjectivity, let $Q\in\mathcal{P}$ have eigenvalues $\mu,\mu^{-1}>0$. Then $c=\frac12(\mu+\mu^{-1})\ge 1$. Put $u=C^{-1}(c)$ and $Y=L(Q)$. This Y is symmetric and traceless. Cayley–Hamilton gives $Q^2=2cQ-I$, so $(Q-cI)^2=(c^2-1)I$. Since $u\,S(u)^2=C(u)^2-1=c^2-1$, we get $Y^2=uI$. Therefore

$$\exp Y=C(u)I+S(u)Y=cI+(Q-cI)=Q.$$ ∎

Proposition 20.11 (Cartan decomposition of $\mathrm{SL}(2,\mathbb{R})$)**.** *The map $K\times\mathfrak{p}\to G$, $(k,X)\mapsto k\exp X$, is a diffeomorphism. Its inverse is*

$$g\mapsto\bigl(g\exp(-X),\,X\bigr),\qquad X=\tfrac12 L(g^{T}g).$$

Proof. Suppose first that $g=k\exp X$ with $k\in K$. Since X is symmetric, so is $\exp X$. Also k is orthogonal. Hence

$$g^{T}g=\exp(X)k^{T}k\exp(X)=\exp(2X).$$

By the exponential lemma, Lemma 20.10, $2X=L(g^{T}g)$, so X and then k are forced. Conversely, given g, the matrix $g^{T}g$ lies in $\mathcal{P}$. Define X and $k=g\exp(-X)$ as stated. Then

$$k^{T}k=\exp(-X)\exp(2X)\exp(-X)=I,\qquad \det k=1,$$

so $k \in K$. The map and its inverse are smooth by the exponential lemma, Lemma 20.10. ∎

Corollary 20.12 ($G = KAK$)**.** *Every $g \in G$ can be written as $g = k_1 a_t k_2$ with $k_1, k_2 \in K$ and $t \geq 0$, and t is unique. The number e^{2t} is the larger eigenvalue of $g^T g$.*

Proof. Write $g = k \exp X$. The symmetric traceless matrix X has eigenvalues $\pm t$ with $t \geq 0$. Take an orthonormal eigenbasis, ordered $(+t, -t)$, and negate one vector if needed. This gives $k' \in K$ with $X = k'(tH)k'^{-1}$. Then $\exp X = k' a_t k'^{-1}$ and $g = (kk')a_t k'^{-1}$. If $g = k_1 a_t k_2$, then $g^T g = k_2^{-1} a_{2t} k_2$ has eigenvalues $e^{\pm 2t}$, which fixes t. ∎

Example 20.13. We compute the Iwasawa, Cartan and KAK data of the lower triangular unipotent matrix

$$g = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

The Cartan data involve the golden ratio $\tau = \frac{1+\sqrt{5}}{2}$. Recall $\tau + \tau^{-1} = \sqrt{5}$ and $\tau - \tau^{-1} = 1$.

Iwasawa. Here $r = \sqrt{2}$, so $k = k_{\pi/4}$, $e^t = \sqrt{2}$ and $x = \tfrac{1}{2}$. Check:

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \sqrt{2} & \frac{\sqrt{2}}{2} \\ 0 & \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Cartan. Here

$$Q = g^T g = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

The half-trace of Q is $c = \frac{3}{2}$. The equation $C(u) = \cosh\sqrt{u} = \frac{3}{2}$ gives $\sqrt{u} = 2\log\tau$, since $\tau^2 + \tau^{-2} = 3$. Also $\sinh\sqrt{u} = \frac{\sqrt{5}}{2}$. Hence

$$X = \frac{1}{2} L(Q) = \frac{2\log\tau}{\sqrt{5}} \begin{pmatrix} \frac{1}{2} & 1 \\ 1 & -\frac{1}{2} \end{pmatrix} = \frac{\log\tau}{\sqrt{5}} \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}.$$

Now $-\det X = (\log\tau)^2$, so the exponential lemma, Lemma 20.10, gives

$$\exp X = \frac{\sqrt{5}}{2} I + \frac{1}{2\sqrt{5}} \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}.$$

Then

$$k = g \exp(-X) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix} \in \mathrm{SO}(2).$$

The eigenvalues of X are $\pm\log\tau$. Hence, as in the proof of the KAK corollary, Corollary 20.12, the KAK parameter of g is $t = \log\tau$.

20.4 The upper half-plane

Let $\mathbb{H} = \{z \in \mathbb{C} : \operatorname{Im} z > 0\}$. For g with entries $\alpha, \beta, \gamma, \delta$ as in the Iwasawa decomposition, Proposition 20.9, put

$$g \cdot z = \frac{\alpha z + \beta}{\gamma z + \delta}.$$

Lemma 20.14. *For $z \in \mathbb{H}$ one has $\gamma z + \delta \neq 0$ and*

$$\operatorname{Im}(g \cdot z) = \frac{\operatorname{Im} z}{|\gamma z + \delta|^2}.$$

This defines an action of G on $\mathbb{H}$. The stabiliser of i is K, and $-I$ acts trivially.

Proof. If $\gamma \neq 0$, the root of $\gamma z + \delta$ is real. If $\gamma = 0$, then $\delta \neq 0$. Next,

$$g \cdot z = \frac{(\alpha z + \beta)(\gamma \bar{z} + \delta)}{|\gamma z + \delta|^2},$$

and the numerator has imaginary part $(\alpha\delta - \beta\gamma) \operatorname{Im} z = \operatorname{Im} z$.

For the action law, write $[v] = v_1/v_2$ for $v \in \mathbb{C}^2$ with $v_2 \neq 0$. Then $g \cdot z = [g(z,1)^T]$, and $[\lambda v] = [v]$ for $\lambda \neq 0$. If h has bottom row (γ', δ'), then $h(z,1)^T = (\gamma' z + \delta')(h \cdot z, 1)^T$. Hence

$$(gh) \cdot z = [gh(z,1)^T] = [g(h \cdot z, 1)^T] = g \cdot (h \cdot z).$$

If $g \cdot i = i$, then $\alpha i + \beta = -\gamma + \delta i$. So $\alpha = \delta$ and $\beta = -\gamma$, with $\alpha^2 + \gamma^2 = 1$. That is, $g = k_\varphi$. Conversely, $k_\varphi \cdot i = ie^{i\varphi}/e^{i\varphi} = i$. Finally, $(-I) \cdot z = z$. ∎

Inverting $g^{-1} = kan$ gives $g = n^{-1}a^{-1}k^{-1}$. So the Iwasawa decomposition, Proposition 20.9, also yields a unique decomposition $G = NAK$.

Proposition 20.15. *Write $g = n_x a_t k$. Then*

$$g \cdot i = x + ie^{2t}, \qquad x = \frac{\alpha\gamma + \beta\delta}{\gamma^2 + \delta^2}, \qquad e^{2t} = \frac{1}{\gamma^2 + \delta^2}.$$

The map $n_x a_t \mapsto n_x a_t \cdot i$ is a diffeomorphism $NA \to \mathbb{H}$. The map $gK \mapsto g \cdot i$ is a bijection $G/K \to \mathbb{H}$.

Proof. We have $a_t \cdot z = e^{2t} z$ and $n_x \cdot z = z + x$. Since $k \cdot i = i$, this gives $n_x a_t k \cdot i = x + ie^{2t}$. The stated formulas for x and e^{2t} follow by comparing real and imaginary parts in the direct computation

$$g \cdot i = \frac{(\alpha i + \beta)(\delta - \gamma i)}{\gamma^2 + \delta^2} = \frac{\alpha\gamma + \beta\delta + i}{\gamma^2 + \delta^2}.$$

The map $(x,t) \mapsto x + ie^{2t}$ has smooth inverse $z \mapsto (\operatorname{Re} z, \tfrac{1}{2} \log \operatorname{Im} z)$. Finally, $g \cdot i = g' \cdot i$ exactly when $g^{-1}g' \in K$, by the stabiliser statement of Lemma 20.14. ∎

So Iwasawa coordinates are Cartesian coordinates (x, y) with $y = e^{2t}$. Here N translates horizontally and A dilates from 0. The subgroup NA acts by the affine maps $z \mapsto e^{2t} z + x$; Exercise 20.6 identifies it with the affine group of Lecture 16.

The KAK form describes $\mathbb{H}$ by K-orbits. If $g \cdot i = z$ and $g = k_1 a_t k_2$, then $z = k_1 \cdot (ie^{2t})$. The number $t \geq 0$ depends only on z, because replacing g by gk does not change the eigenvalues of $g^T g$. So $\mathbb{H}$ is the disjoint union of the K-orbits of the points ie^{2t}, $t \geq 0$. Exercise 20.5 shows that these orbits are Euclidean circles.

20.5 Self-checks

1. What are the real ranks of $\mathfrak{sl}(n,\mathbb{R})$ and $\mathfrak{su}(n)$? What do A and N become for $\mathrm{SU}(n)$?
   Answer. The real rank of $\mathfrak{sl}(n,\mathbb{R})$ is $n-1$, and the real rank of $\mathfrak{su}(n)$ is 0. For $\mathrm{SU}(n)$, $\mathfrak{p}=0$, so $A=N=\{I\}$ and the Iwasawa decomposition reads $G=K$.
2. Find the Iwasawa decomposition of $g=\begin{pmatrix}0&1\\-1&2\end{pmatrix}$.
   Answer. Here $r=1$, $\cos\varphi=0$ and $\sin\varphi=-1$, so $k=k_{-\pi/2}$ and $a=I$. Also $x=(0\cdot 1+(-1)\cdot 2)/1=-2$. Indeed
   $$\begin{pmatrix}0&1\\-1&0\end{pmatrix}\begin{pmatrix}1&-2\\0&1\end{pmatrix}=\begin{pmatrix}0&1\\-1&2\end{pmatrix}.$$
3. Where does $-I$ sit in $G=KAN$, and what does it do on $\mathbb{H}$?
   Answer. In $G=KAN$, the element $-I$ has factors $k=k_{\pi}$ and $a=n=I$. The element $-I$ acts trivially on $\mathbb{H}$. Hence the action factors through $G/\{\pm I\}$.

Summary

A Cartan involution splits $\mathfrak{g}=\mathfrak{k}\oplus\mathfrak{p}$. The compact form $\mathfrak{k}\oplus i\mathfrak{p}$ of $\mathfrak{g}_{\mathbb{C}}$ has real rank zero. The split form $\mathfrak{sl}(n,\mathbb{R})$ has real rank $n-1$. For connected semisimple groups with finite centre, the cited Cartan and Iwasawa theorem gives $G=K\exp\mathfrak{p}$ and $G=KAN$. For $\mathrm{SL}(2,\mathbb{R})$ we proved both decompositions, with explicit inverses. On $\mathbb{H}$, identified with G/K, the subgroup NA gives Cartesian coordinates. The KAK form describes $\mathbb{H}$ as the disjoint union of the K-orbits of the points ie^{2t} with $t\geq 0$.

Reading

- Knapp Beyond, Chapter VI, Sections 1–4: compact real forms, the Cartan decomposition on the Lie algebra and group levels, and the Iwasawa decomposition.
- P. Etingof, *Lie Groups and Lie Algebras*, MIT 18.755 lecture notes, Lectures 15 and 18: a second route through real forms and these decompositions.
- Lecture 22 uses $G=KAN$ to construct the principal series.